
BioTrack Studio

Illustrated User Guide

A no-code workflow for cell segmentation, tracking, consensus-assisted review, and manual correction.

START HERE: Extract the complete ZIP, read this guide, and test the included demo_data before using experimental images.

Windows	macOS	Linux
First use: setup_conda.bat Then: winstart.bat	First use: bash setup_conda.sh Then: bash start.sh	First use: bash setup_conda.sh Then: bash start.sh

Consensus markers show where algorithms disagree. They do not prove that a result is wrong. The reviewer makes the final decision from the microscopy image and biological context.

Research software. Not intended for clinical diagnosis.

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For: wet-laboratory researchers and imaging scientists.

Use this protocol to install BioTrack Studio, run one complete analysis, review consensus markers, save corrections, and export a review report. The illustrated PDF is the main document for non-programmers.

Use the notebook `BioTrack_Studio_Demo.ipynb` only when you need the full reference: all method choices, input rules, consensus parameters, server notes, CSV columns, scientific limitations, and publication records. No notebook cell must be executed to use the software.

> Consensus means that two algorithms are compared. A marker shows where they > disagree; it does not prove that either result is wrong. Always decide from > the raw image and biological context.

1. Prepare the Images

Choose one folder containing a single-channel 2D time series. It may contain:

- one TIFF file per time point; or
- one multi-page TIFF whose pages are consecutive time points.

For separate files, make sure the filenames sort in time order. Names such as `t000.tif`, `t001.tif`, and `t002.tif` are recommended. All frames must have the same width, height, and data type. Do not mix unrelated TIFF files in the input folder.

Create a new empty output folder for each analysis. Keep the original images in a separate location.

Included demo

Run the included `demo_data` before using experimental data. It contains 20 consecutive frames of real HeLa cells expressing H2B-mCherry. The sequence was cropped uniformly from a CC BY 4.0 dataset by Romain Guet; full attribution and processing details are in `demo_data/DATA_SOURCE.txt`.

The demo checks installation and shows how consensus works. It is not ground truth and is not an accuracy benchmark.

2. Install and Open

Extract the complete release ZIP first. Do not run a launcher from inside the ZIP archive.

Windows

1. On the first use, double-click `setup_conda.bat` and wait for it to finish.
2. Double-click `winstart.bat`.
3. For later sessions, use `winstart.bat` directly.

macOS

1. Open Terminal.
2. Type `cd` followed by one space. Do not press Return yet.
3. Drag the extracted `BioTrack_Studio` folder from Finder into Terminal.
4. Press Return.
5. On the first use, run:

```
bash setup_conda.sh
```

6. Start BioTrack Studio with:

```
bash start.sh
```

For later sessions, return to the folder and run only `bash start.sh`.

Linux

Open a terminal in the extracted folder and run:

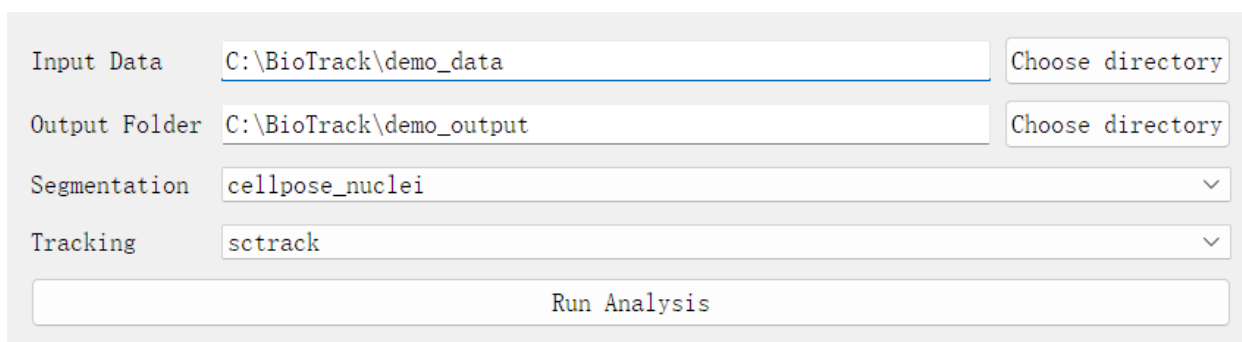
```
bash setup_conda.sh
bash start.sh
```

For a remote Linux server, napari needs a graphical display. Connect with X11 forwarding before launching. MobaXterm can provide the display when X11-Forwarding is enabled in the SSH session.

The first setup may take tens of minutes because several isolated environments are created. Keep the launcher open. If setup is interrupted, start the same launcher again; completed steps are reused.

3. Run the Demo Pipeline

In napari, open Plugins > BioTrack Studio > Cell Analysis Pipeline.



Cell Analysis Pipeline

1. For Input Data, select the included demo_data folder.
2. For Output Folder, select a new empty folder.
3. Select cellpose_nuclei for Segmentation.
4. Select sctrack for Tracking.
5. Select Run Analysis and wait until the result opens in napari.
6. Move through the first, middle, and final frames.
7. Confirm that colored masks follow the visible nuclei and tracks extend over several frames.

A complete run creates:

File	Meaning
image.tif	Image sequence used by the pipeline
mask.tif	Labelled segmentation for every frame
track.csv	Track identity, frame, state, mask label, and position
config.yaml	File and column mapping for correction tools

The default cell state is none unless a tracker supplies another state.

4. Analyze Your Own Data

Use the same Pipeline panel and select the method for the object visible in your images.

Segmentation	Intended starting point
Cellpose Cyto	Whole cells or cytoplasm-like objects
Cellpose Nuclei	Fluorescent nuclei
StarDist	Round or star-convex fluorescent nuclei

Tracking	Intended starting point
SC-Track	Longer sequences and lineage-aware tracking
TrackPy	Clearly separated objects
Ultrac	Crowded or ambiguous scenes

SC-Track requires at least 13 frames. No model works best for every specimen. Inspect the mask before interpreting tracks. Do not compare a whole-cell mask with a nuclear mask as though they represented the same object.

5. Open Manual Correction

Open Plugins > BioTrack Studio > Manual Correction.

Result Folder

C:\BioTrack\Result

Choose directory

Correction Tool

MMV_H4Tracks

☒ Tracking consensus assistance

Primary Tracking Result

Auto-detect from result folder

☒ Mask consensus assistance

Mask Reference

Automatic

Mask Settings

Automatic

Open Correction Workspace

Manual Correction options

1. Select the complete result folder.
2. Select napari-amdtrk or MMV_H4Tracks.
3. Leave both consensus boxes clear for ordinary correction, or enable one or both boxes for consensus assistance.
4. Select Open Correction Workspace.

Use napari-amdtrk when the corrected result must remain in BioTrack Studio's standard mask.tif and track.csv format. Its Save button updates those files. MMV_H4Tracks provides its own tab-based tools and saves directly loaded BioTrack layers to a separate Zarr file; it does not replace the standard TIFF and CSV files.

Official correction tutorials: [napari-amdtrk guide](#), [napari-amdtrk video](#), and [MMV_H4Tracks guide](#).

BioTrack Studio installs and opens both tools automatically, so do not repeat the installation or file-loading steps shown there. The external pages may show a newer interface; follow this guide for opening files, consensus assistance, and saving BioTrack results.

6. Use Consensus Assistance

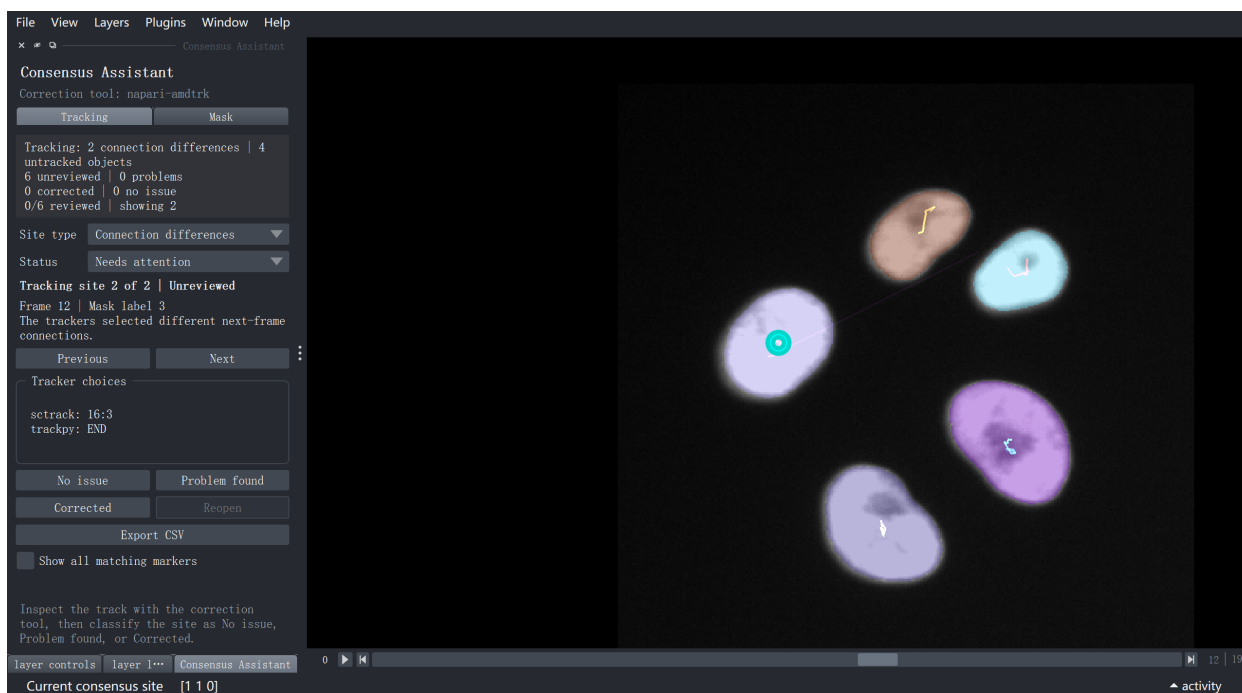
For the included demo:

1. Select its completed result folder.
2. Select napari-amdtrk.
3. Enable Tracking consensus assistance.
4. Enable Mask consensus assistance.
5. Keep the primary tracker, mask reference, and mask settings on Automatic.
6. Select Open Correction Workspace and wait while the comparison runs.

The original result remains unchanged. BioTrack Studio creates a dated copy in:

```
consensus_assist/runs/<date_time>/correction
```

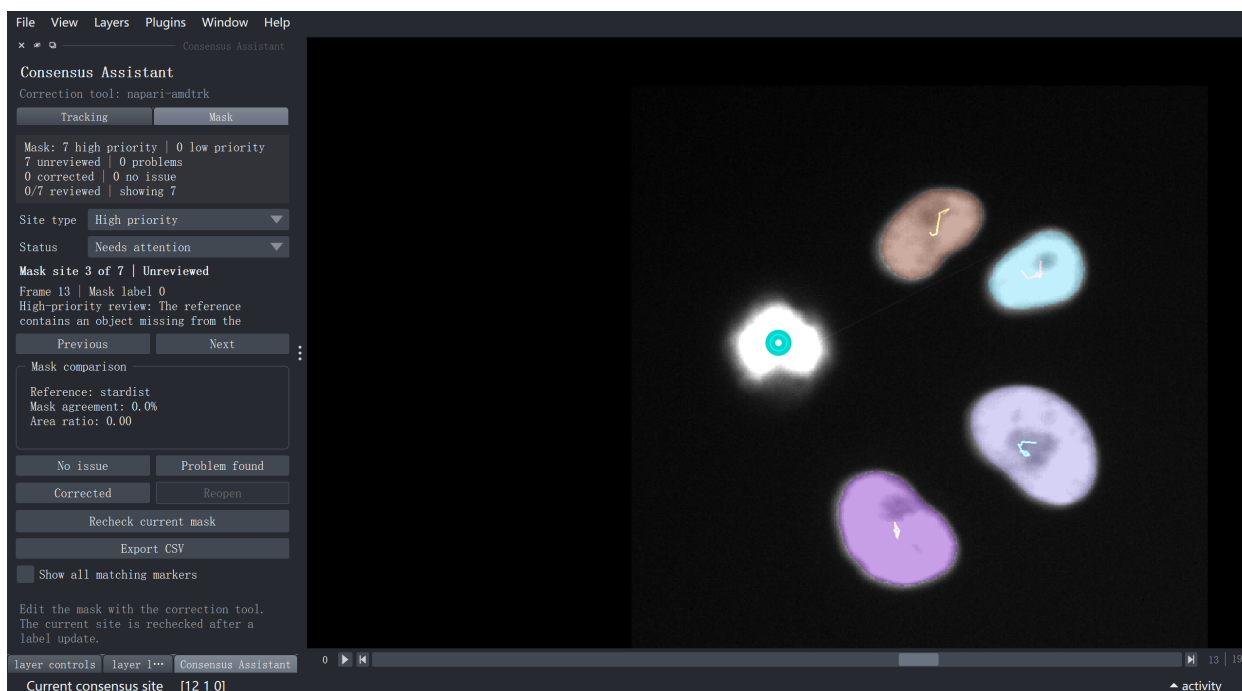
The tested Windows reference run produced six tracking review sites: two connection differences and four segmented objects missing from the SC-Track table. Small numerical differences are possible across computers.



Tracking consensus review

The Tracking tab shows the current frame, mask label, and the next-frame choice made by SC-Track and TrackPy. Start with Connection differences, then inspect Untracked objects. The cyan ring marks the current site.

The same reference run produced seven high-priority mask sites and no low-priority sites. All seven were visible nuclei present in StarDist but absent from the Cellpose Nuclei mask. Several are the same biological cell at different frames, so these are seven frame-level review sites, not seven unique cells.



Mask consensus review

The Mask tab compares Cellpose Nuclei with StarDist. High priority means a strong model disagreement; it still requires human review. The reference mask is not ground truth.

7. Review, Correct, and Save

For each marker:

1. Inspect the raw image, mask, and local track.
2. Use Previous and Next to move through the filtered queue.
3. Choose one classification:

Button	Meaning
No issue	The primary result is acceptable
Problem found	A real problem exists but is not corrected yet
Corrected	The edit was completed and saved
Reopen	Return a previous decision to the review queue

For a real problem, use this order:

1. Select Problem found.
2. Correct the mask or track with the correction tool.
3. Select Save in napari-amdtk.
4. Return to the Consensus Assistant and select Corrected.

Selecting Corrected records the review decision; it does not edit or save a mask by itself. After editing a mask, Recheck current mask tests whether the local disagreement remains in the open napari layer.

8. Export the Review Report

Select Export CSV. The assisted correction folder contains:

File	Use
consensus_review_status.csv	Saves classifications when the review is reopened
consensus_review_report.csv	One row for every review site
consensus_review_summary.csv	Overall, Tracking, and Mask totals

These files open directly in Excel. Report initial sites, confirmed problems, corrected problems, accepted sites, and sites still needing attention as separate values. A decrease in open markers is review progress, not an accuracy score.

9. Finish and Check

Before closing the correction window, confirm that:

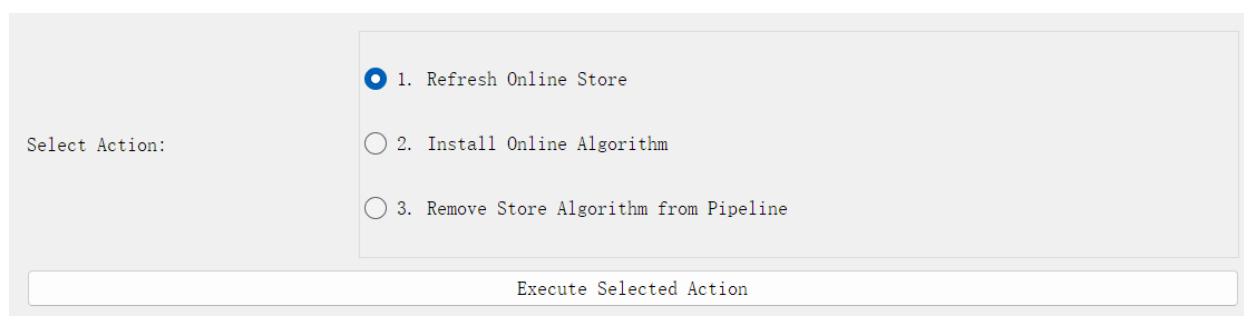
- early, middle, and final frames display correctly;
- corrections were saved in the intended format;
- the review CSV files were exported;
- the original Pipeline result was kept; and
- the selected algorithms and any custom settings were recorded.

For complete input rules, skip modes, custom mask-consensus parameters, CSV columns, server instructions, publication records, and scientific limitations, open `BioTrack_Studio_Demo.ipynb` in the docs folder.

10. Add a Tested Algorithm

The optional Algorithm Store installs additional methods that have been adapted and tested for BioTrack Studio. It is not required for Cellpose, StarDist, SC-Track, TrackPy, or Ultrack.

Open Plugins > BioTrack Studio > Algorithm Store.



Select Action:

- ☒ 1. Refresh Online Store
- ☐ 2. Install Online Algorithm
- ☐ 3. Remove Store Algorithm from Pipeline

Execute Selected Action

Algorithm Store

1. Select Refresh Online Store, then select Execute Selected Action.
2. Select Install Online Algorithm.
3. Choose an available algorithm. An empty list means that no additional validated method is currently published.

-
4. Select Execute Selected Action and keep napari open while the isolated environment is installed.
 5. Wait for the successful-installation message.
 6. Close and reopen Cell Analysis Pipeline. The new method will appear in the appropriate Segmentation or Tracking list.
 7. Run the included demo or the test data supplied with the adapter before analyzing experimental data.

To hide a Store-installed method, choose Remove Store Algorithm from Pipeline. This removes its menu registration but deliberately keeps the conda environment and downloaded files so that a failed or accidental removal does not destroy the installation. It therefore does not free disk space. Included BioTrack Studio methods cannot be removed or replaced through the Store.

The Store does not make arbitrary GitHub code compatible. Every listed method must provide a BioTrack Studio adapter that accepts the standard inputs and writes `mask.tif` or `track.csv`. Treat a newly installed algorithm as a new scientific method: check its license and citation, record its version and settings, and validate its output for the specimen being studied.

Help, License, and Citation

If the demo fails, retry the same launcher once. If it still fails, keep the installation log and report the operating system, selected action, last visible message, and whether napari opened. Do not upload private microscopy data or credentials.

BioTrack Studio is research software and is not intended for clinical diagnosis. It is distributed under GNU GPL version 3. Integrated tools and demo data retain their own licenses. See `LICENSE`, `THIRD_PARTY_NOTICES.md`, `ACKNOWLEDGEMENTS.md`, and `demo_data/DATA_SOURCE.txt`. Cite the original segmentation, tracking, correction, and dataset publications used in an analysis.